PORTFOLIO ISSUE — 02 // MASTER OF COMPUTER APPLICATIONS (MCA) PROJECT REPORT

HARSH RATHORE — CREATIVE PORTFOLIO

Full-Stack MERN Architecture, Autonomous LLM Pipelines & Ultra-Luxury Fluid Web Interfaces

DEVELOPER: Harsh Rathore **DEGREE:** Master of Computer Applications (MCA) **LOCATION:** Surat, Gujarat, IN (IST) **DEPLOYMENT:** Vercel / Netlify

A // PORTFOLIO OVERVIEW & ARCHITECTURAL MANIFESTO

EXECUTIVE SUMMARY

Project Synopsis: The portfolio is an ultra-luxury, high-performance personal engineering showcase engineered at the intersection of haute-couture editorial typography, brutalist structural forms, and next-generation fluid interface physics (inspired by **ColorOS Aqua Dynamics & iOS Liquid Glass**). Handcrafted with raw, zero-dependency **HTML5, CSS3, and ES6+ Vanilla JavaScript**, it deliberately rejects bloated generic templates in favor of absolute performance, strict accessibility, and micro-engineered kinetic fidelity.

Core Academic & Professional Purpose: Built as an exhaustive technical repository for formal MCA submission and production engineering evaluation. The portfolio demonstrates how thoughtful human interfaces converge with resilient, scalable engineering across the entire software lifecycle—leveraging **React** for reactive UI states, **Node.js & Express** for high-throughput REST APIs and middleware pipelines, **MongoDB Atlas** for document persistence and vector search, and **Google Gemini API** for autonomous AI tool orchestration.

Production Deployment & Verified Access: Optimized for zero-configuration edge delivery. Configured for Vercel production hosting via **harsh-pratapsingh-portfolio.vercel.app** with complete source code archived at **github.com/harsh-pratap-singh-rathore**.

B // VERIFIED TECHNOLOGY STACK & ARCHITECTURAL ECOSYSTEM

SOURCE INSPECTED

DOMAIN / LAYER	TECHNOLOGIES, FRAMEWORKS & LIBRARIES	ARCHITECTURAL ROLE & IMPLEMENTATION CONTEXT
Core Languages	JavaScript (ES6+), TypeScript, HTML5, CSS3, PHP 8.x, Python 3.x	Isomorphic client/server scripting, type-safe API routing, and AI data manipulation.
Frontend & UI	Vanilla JS Engine, React 19, CSS3 Design Tokens, Tailwind CSS, Vite	Zero-runtime client DOM engine, fluid viewport scaling, glassmorphic optics.
Backend & APIs	Node.js, Express.js, RESTful Architecture, JWT Auth, CORS, SSE Streams	Event-driven asynchronous I/O, modular middleware pipelines, and secure session management.
Databases & Caching	MongoDB Atlas, Mongoose, MySQL, PostgreSQL, Redis In-Memory Cache	Document storage, relational civic reporting schemas, and sub-50ms query caching.
AI Systems & LLMs	Google Gemini API (2.5 Flash), Ollama Runtime, PyTorch, HNSW Vector RAG	Autonomous tool-calling agents, dense vector semantic search, and hybrid BM25 retrieval.
DevOps & Infrastructure	Vercel, Netlify, Git, GitHub Actions, Docker Containers, Linux / Unix CLI	Edge routing, continuous integration, containerization, and static asset distribution.

C // SYSTEM ARCHITECTURE & KEY FEATURES (PART 1: INTERACTIVE ENGINES)

ENGINEERING RIGOR

1. Cinematic Preloader & Session Cache	2. Fluid Liquid Glass Dual Navigation	3. Real-Time Telemetry & Optics
• **Accelerated Entry:** 1.25s requestAnimationFrame 00–100% counter with 130ms letter reveal. • **Session Caching:** sessionStorage checks bypass preloader in 40ms on repeat visits. • **A11y Compliant:** Applies inert focus trapping to main DOM during intro execution.	• **Desktop Segmented Glass:** Capsule bar with spring-overshoot elasticity (scaleX(1.24)). • **Mobile Glass Dock:** Ergonomic iOS floating bottom dock (#iosBottomDock) with touch haptics. • **Kinetic ScrollSpy:** Dynamic capsule glide tracking active section viewports.	• **Live IST HUD Clock:** Real-time Indian Standard Time (GMT+05:30) updated dynamically. • **Magnetic Dual Cursor:** Interpolated dot & ring with magnetic button attraction. • **GPU Optimization:** Specular 3D hover effects throttled via requestAnimationFrame.

C.1 // ACTIVE ARCHITECTURAL CYCLES & SKILL PROGRESS TELEMETRY

VERIFIED BENCHMARKS

JAVASCRIPT (ES6+): 90% MASTERY
Async patterns, Event loop mechanics, Closures, DOM engine architecture, Prototypes

MERN ECOSYSTEM: 82% MASTERY
React 19 hooks, Express modular routing, MongoDB document modeling & Atlas vector search

DL · ML · OLLAMA: 78% MASTERY
Deep learning neural nets, Scikit algorithms, Ollama local model inference, PyTorch pipelines

APIS & DATA SYSTEMS: 75% MASTERY
RESTful API design, JWT/Session authentication, MySQL relational schemas & Redis cache

C.2 // FEATURED PROJECTS & ACADEMIC RESEARCH MONOGRAPHS

CONFIRMED IMPLEMENTATIONS

PROJECT / SPECIFICATION	CORE ARCHITECTURE & PROBLEM SOLVED	STACK & VERIFIED ACCESS LINKS
01 // Dropify (Flagship) Audiophile Music Player	Next-gen audiophile music streaming platform featuring a custom 10-band Web Audio DSP equalizer, dynamic stereo spatializer, LRCLIB synchronized lyrics engine, YouTube stream decoding via InnerTube, and cross-platform clients (Web, Android APK v1.0.2, Electron).	**Stack:** React 18, Web Audio DSP, Node.js, Kotlin **Live URL:** dropify.vercel.app [Visit] **Repository:** Dropify Repo [Code]
02 // Shree Karni Textiles B2B Wholesale Commerce	High-throughput enterprise fabric procurement portal indexing 10,000+ m² inventory. Engineered with multi-layered Redis caching, dynamic catalog indexing, and low-latency inquiry dispatch (48ms TTFB).	**Stack:** PHP, JS, CSS3, Composer, HTML5 **Live URL:** shreekarnitextiles.com [Visit]
03 // Code Canvas Developer Tooling / IDE	Real-time interactive CSS design studio and component workbench. Enables developers to manipulate box models, gradients, and typography directly on canvas with instant compiled CSS extraction.	**Stack:** JavaScript, HTML5, CSS3, PHP **Live URL:** codecanvas.page [Visit] **Repository:** CodeCanvas Repo [Code]
04 // My Voice Civic Infrastructure	Community public infrastructure incident triage and dispatch network. Features GPS-tagged incident logging, role-based admin triage, and relational status tracking.	**Stack:** PHP, MySQL, JavaScript, CSS3 **Live URL:** myvoice.gt.tc [Visit]
Research Monographs Editorial AI Archive	• **Autonomous LLM Agents:** Deterministic state machine bounds preventing runaway loops. • **Dense Vector Retrieval:** Hybrid BM25 + dense vector embeddings in MongoDB Atlas. • **Resilient Async Pipelines:** Transactional rollback state bus resolving race conditions.	**Domain:** LLMs · Vector Search · RAG **Archive:** github.com/harsh-pratap-singh-rathore

D // AI TOOLS UTILIZED — GOOGLE GEMINI IN ENGINEERING LIFECYCLE

DEVELOPMENT ASSISTANCE

Explicit Attribution: Google Gemini served as the primary AI development assistant throughout the engineering and design lifecycle of this portfolio project. Gemini was utilized deliberately as a high-precision pair-programming and architectural reasoning companion across four primary phases:

• **Vanilla JavaScript Motion & Physics Guidance:** Gemini assisted in drafting and mathematically tuning the custom kinetic algorithms—including the spring-overshoot stretching physics (cubic-bezier(0.34, 1.56, 0.64, 1)) for the segmented glass navigation capsules, the 3D card stack swipe geometry, and the interpolated magnetic cursor engine.

• **Glassmorphism & Haute-Couture UI/UX Refinement:** Formulated multi-layered liquid acrylic styling rules (backdrop-filter: blur(44px) saturate(220%)), surface-tension borders, and responsive CSS Grid/Flexbox architectures.

• **Performance Profiling & Layout Optimization:** Guided network payload optimization (reducing initial asset load by **2.05 MB** by replacing uncompressed PNGs with an 86.5 KB optimized fallback portrait), eliminating layout thrashing via requestAnimationFrame throttling, and implementing sessionStorage preloader caching.

• **Security Hardening & Deployment Engineering:** Aided in structuring enterprise-grade HTTP security headers inside vercel.json to enforce MIME sniffing prevention, clickjacking protection, and static cache revalidation.

E // VERIFIED PROFESSIONAL NODES & CONTACT CHANNELS

AUTHENTICATED PROFILES

CHANNEL / PLATFORM	CONFIGURED IDENTIFIER / HANDLE	CLICKABLE HYPERLINK & ACCESS NODE
GitHub Source Archive	harsh-pratap-singh-rathore (20+ Repos)	https://github.com/harsh-pratap-singh-rathore
LinkedIn Professional Graph	harsh-rathore-1597012aa (Verified Profile)	https://www.linkedin.com/in/harsh-rathore-1597012aa/
Direct Dispatch Email	harshpratapsinghrathore555@gmail.com	mailto:harshpratapsinghrathore555@gmail.com
Dropify Repository	Dropify Flagship Audiophile Repo	https://github.com/harsh-pratap-singh-rathore/Dropify
Code Canvas Repository	CodeCanvas UI Workbench Repository	https://github.com/harsh-pratap-singh-rathore/CodeCanvas

F // DEVELOPMENT WORKFLOW, CODEBASE STRUCTURE & DEPLOYMENT

PRODUCTION CONFIG

Codebase Architecture: Structured with strict separation of concerns and zero build-step overhead: index.html (semantic markup with full WAI-ARIA accessibility attributes), style.css (5,400+ lines of custom tokens, fluid typography with CSS clamp(), and liquid acrylic optics), script.js (modular uncompiled vanilla JS engine driving preloader, cursor, docks, and 3D card stacks), and data.js (single-source-of-truth data model for projects, research, and profile credentials).

Production Deployment Pipeline: Optimized for instant, zero-configuration deployment on **Vercel** and **Netlify**. Configured via vercel.json with enterprise security headers: X-Content-Type-Options: nosniff (MIME protection), X-Frame-Options: DENY (clickjacking prevention), and X-XSS-Protection: 1; mode=block. Static assets feature strict revalidation headers (public, max-age=0, must-revalidate) ensuring rapid client delivery and instant rollouts.

CANDIDATE DECLARATION & SUBMISSION NOTE:
This report is submitted in partial fulfillment of the requirements for the Master of Computer Applications (MCA) evaluation. All technologies, features, and source repositories documented herein have been rigorously inspected and verified.

CANDIDATE SIGNATURE:
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